Code No: 136AQ **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
COMPILER DESIGN
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a) Define finite automata. [2]
b) Differentiate pass and phase. [3]
c) If I is set of items of LR parser, the define CLOSURE(I). [2]
d) Differentiate top-down parsing and bottom-up parsing. [3]
e) Give different variants of syntax tree. [2]
f) What is the purpose of intermediate code in the context of compiler design? [3]
g) Define a flow graph. Give an example for it. [2]
h) Define basic block. [3]
i) What is the meaning of reduction in strength optimization? [2]
j) What do you mean by machine independent optimization? How it is different from machine dependent optimization? [3]

PART – B

(50 Marks)

2.a) Obtain Finite Automata for the regular expression **0*(11)* 0.**
b) Explain the functions of lexical analyzer. [5+5]

OR

3.a) Design finite automata for the regular expression **(a+b)*aab***.
b) Explain the concept of boot strapping to generate new compiler from existing compilers. [5+5]

4. Construct SLR parsing table for the following context free grammar:
E → E + T | T
T → T * F | F
F → id
Where **V = { E, T, F }, T = { +, *, id }** and **E** is start variable. [10]

OR

5. Construct LL parsing table for the following context free grammar.
S→ CC
C→aC|b|a
Where **V = {S,C}, T = {a,b}** and **S** is start variable. [10]

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6.a) Write intermediate code for the expression **a=b*-d + c*-d**.

b) What is type checking? Give the rules for type checking of control flow and switch statements. [5+5]

OR

7.a) Explain different Syntax-Directed Translation Schemes.

b) What are different forms of intermediate code? [5+5]

8.a) Explain Peephole optimization.

b) Describe the issues in the design of code generator. [5+5]

OR

9.a) Explain stack allocation method for runtime memory.

b) Explain how to optimize basic block. [5+5]

10.a) Explain the following optimization techniques:
i) Redundant sub expression elimination
ii) Copy propagation

b) How flow graph is useful in optimization? [7+3]

OR

11.a) Explain the following optimization techniques.
i) Loop optimization
ii) Dead-code elimination.

b) Explain data flow analysis procedure with suitable example. [5+5]

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